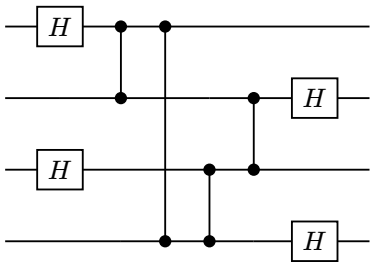
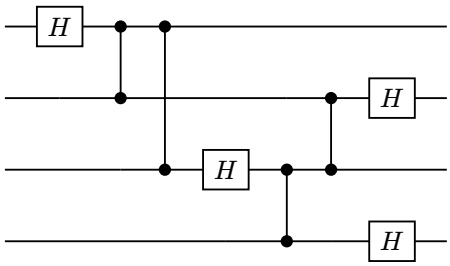
SINV circuit

we proof that a circuit with the following construction is shift invariant:





The feature of this cirucit is:

- there is only one Hadamard gate on each qubit
- control phase gate $\mathrm{CP}(i,j)$ only appear once for each (i,j)
- for each CP gate, there are one H on left and one H on right

proof sketch

the circuit with above structure will map $|b\rangle$ to

$$|b_1 b_2 ...\rangle \to \sum_{c_i \in \{0,1\}} \exp\left(\sum_{j,k} \theta_{j,k} b_j c_k\right) |c_1 c_2 ...\rangle$$

when we consider group $\mathbb{Z}_{2^n}$, the subgroup obtain

$$H = \{b_1 b_2 ... b_m 0 ... 0 \mid b_i \in \{0,1\}\}$$

and the coset is

$$c + H = \{b_1 b_2 ... b_m b'_{m+1} ... b'_n \mid b_i \in \{0,1\}\}$$

thus,

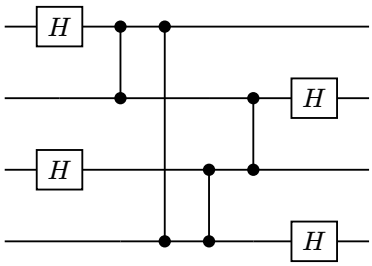
$$\langle g|U|c+H\rangle = \exp\left(\sum_{j,k} \theta_{j,k} b'_j c_k\right) \sum_{b_i} \exp\left(\sum_k \sum_{1 \le j \le m} \theta_{j,k} b_j c_k\right)$$

things that interesting

Its not only QFT is SINV, but the above circuit.

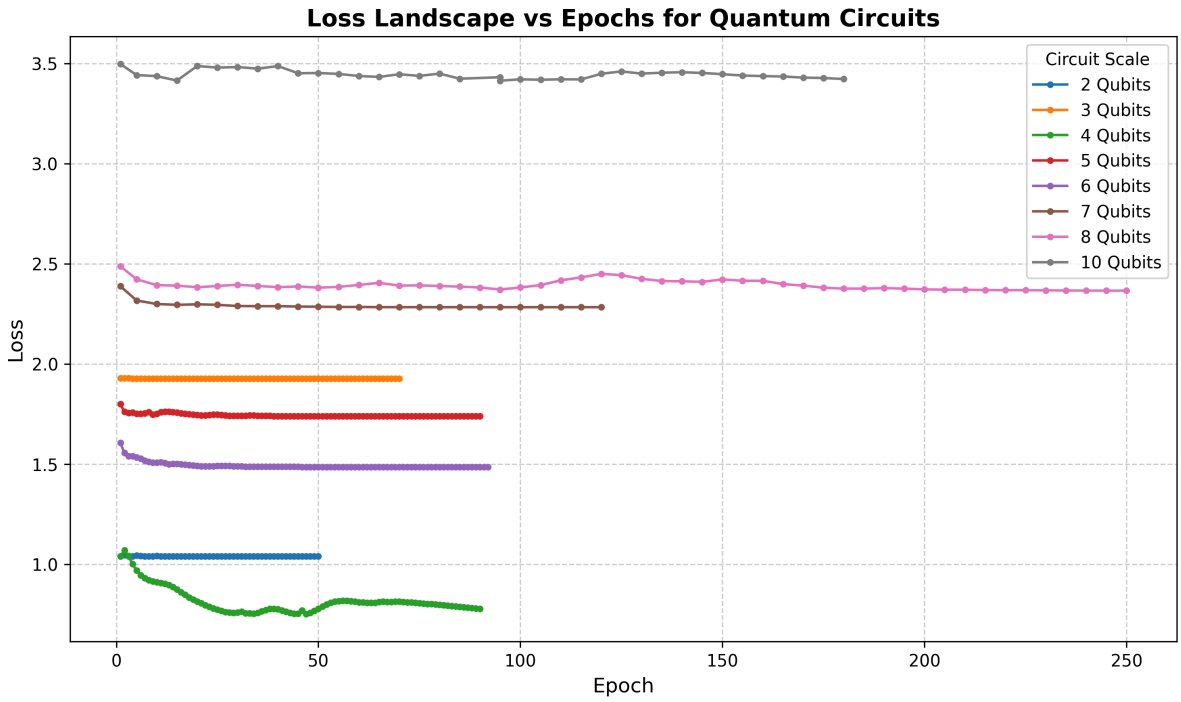
Fisher Information

now we consider circuit with ansatz



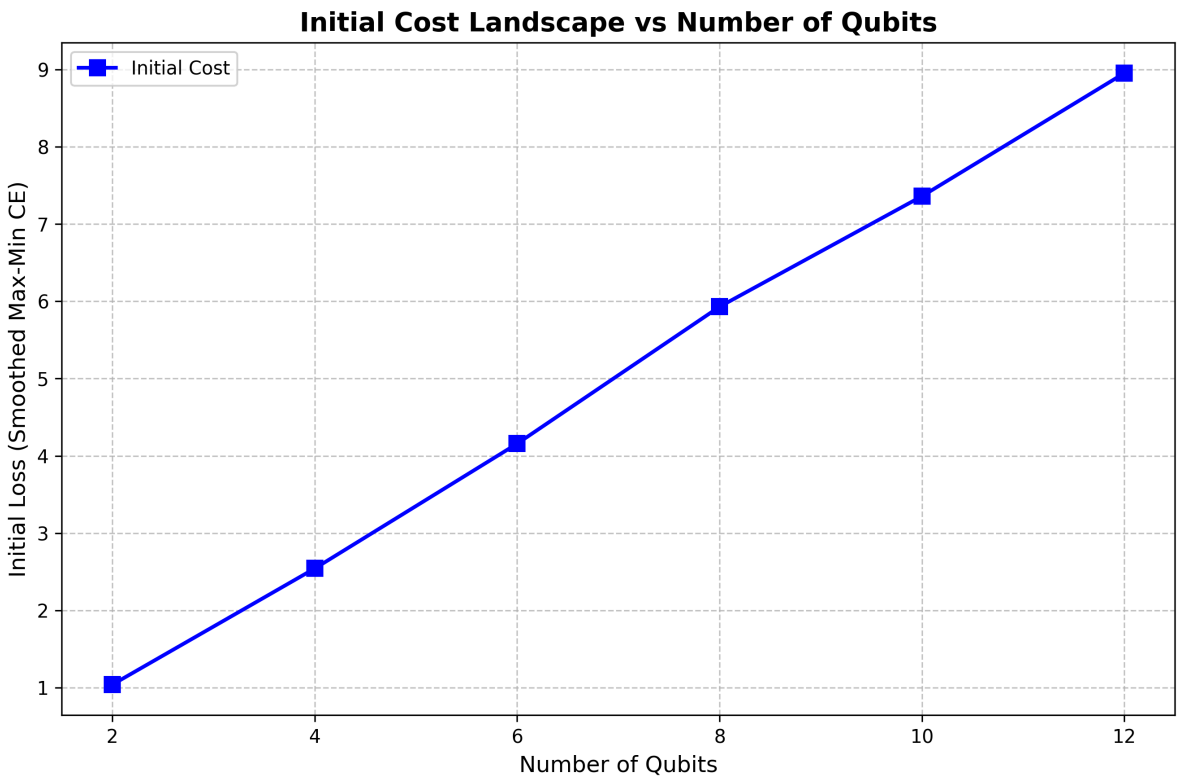
The interlace hih circuit.

we let the phase be $\frac{\pi}{2^{\mathrm{d}}} + \theta$, and we train the parameter θ. We have



the cost function is $-$FI. So the NN aims to maximize the fisher info. The result shows that, the FI does not decrease a lot in training.

So we consider the initial parameter as a relatively good choice of parameter for FI. And we care about how FI goes up when qubits goes up. The numerical result shows that:

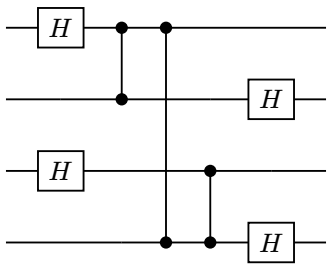


Surprising! It seems that the FI goes linearly!

It seems that the new circuit is tentitive of period!

A difference with former result

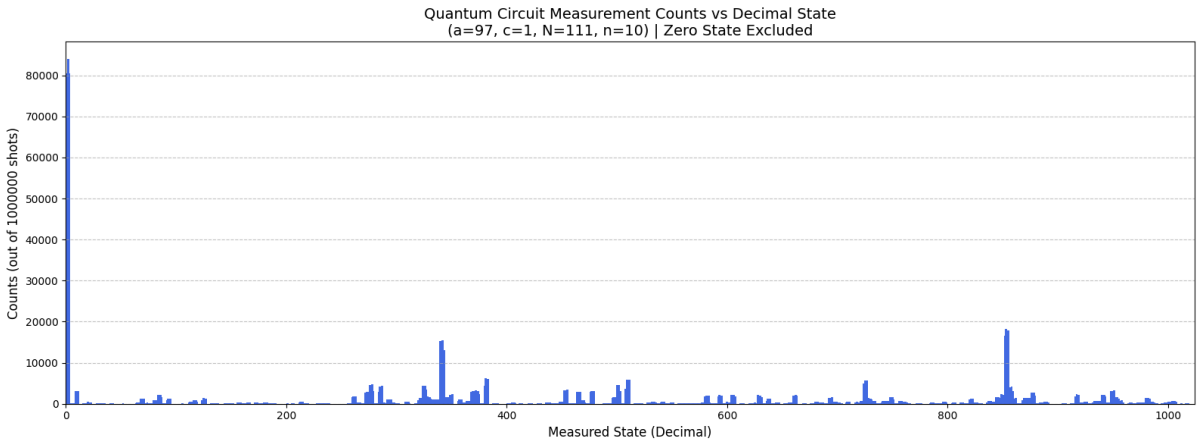
In former result, the circuit is



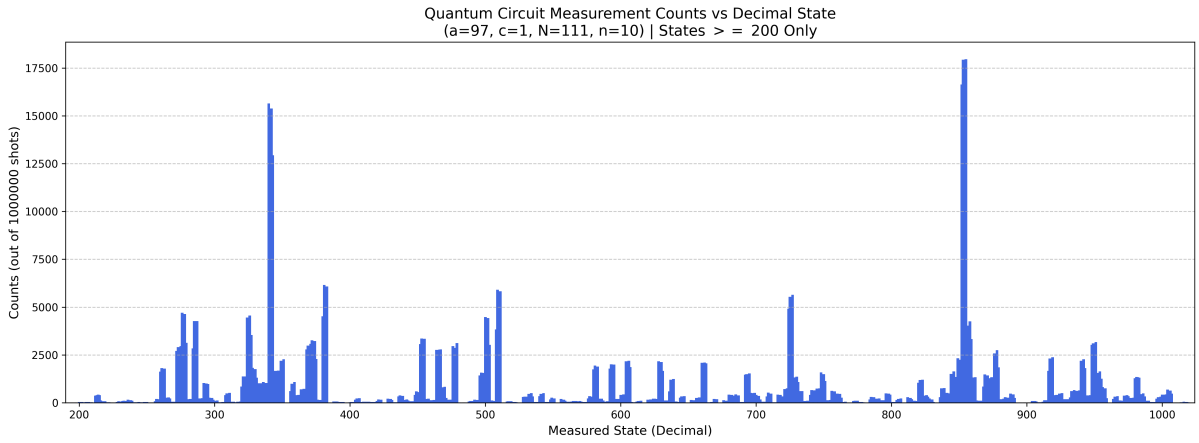
The key thing is, the first qubit do not have any info about the third qubit! I guess it is a big reason why the FI do not perform well on that ansatz.

Machine learning

we apply U on state $|c + H\rangle$, and measure it $n^2 * 1024$ times. the result is



we can zoom the non-zero area



you can see that the frequency carries information, since it is not “uniform”.

so, I use the frequency and a and N to train the NN

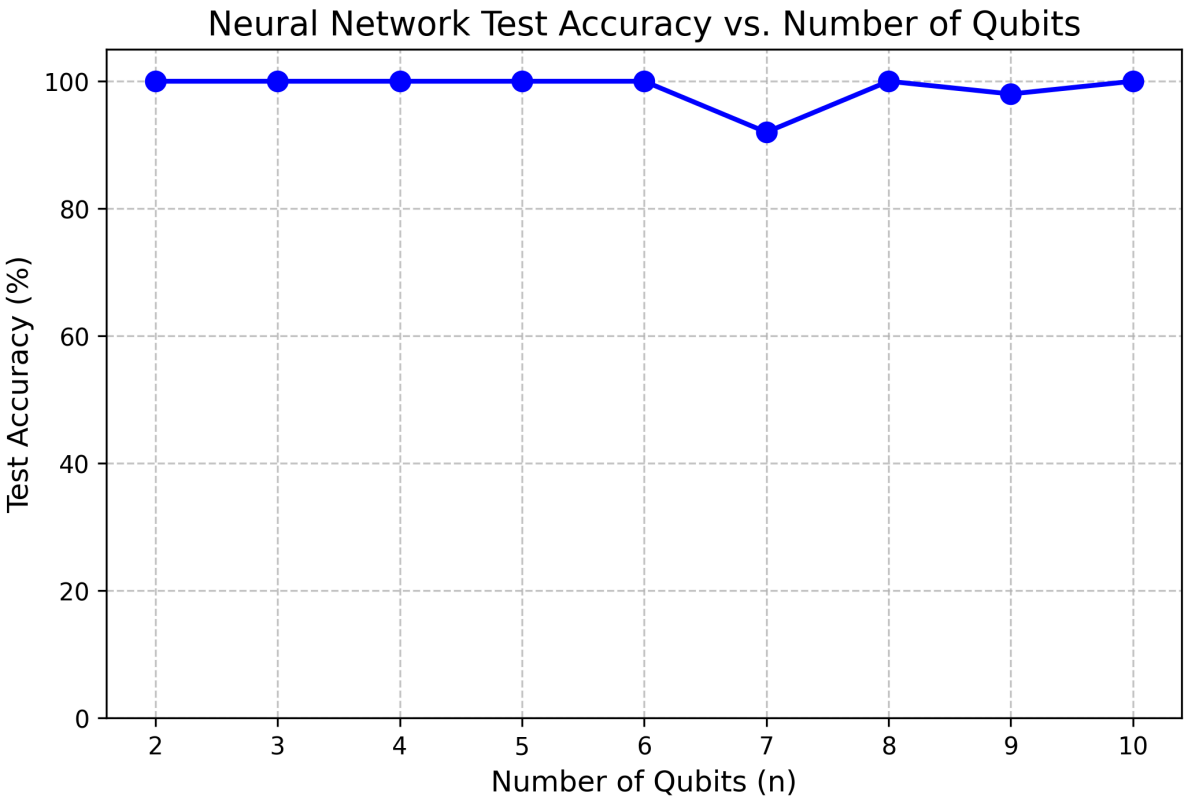
NN set-up

- data: x = (counts, a, N)
- target: y = r s.t. $a^r = 1 \bmod N$.
- nn: $z_{i+1} = \text{relu}(W_i z_i)$
- cost: cross entropy with real r

NN problem

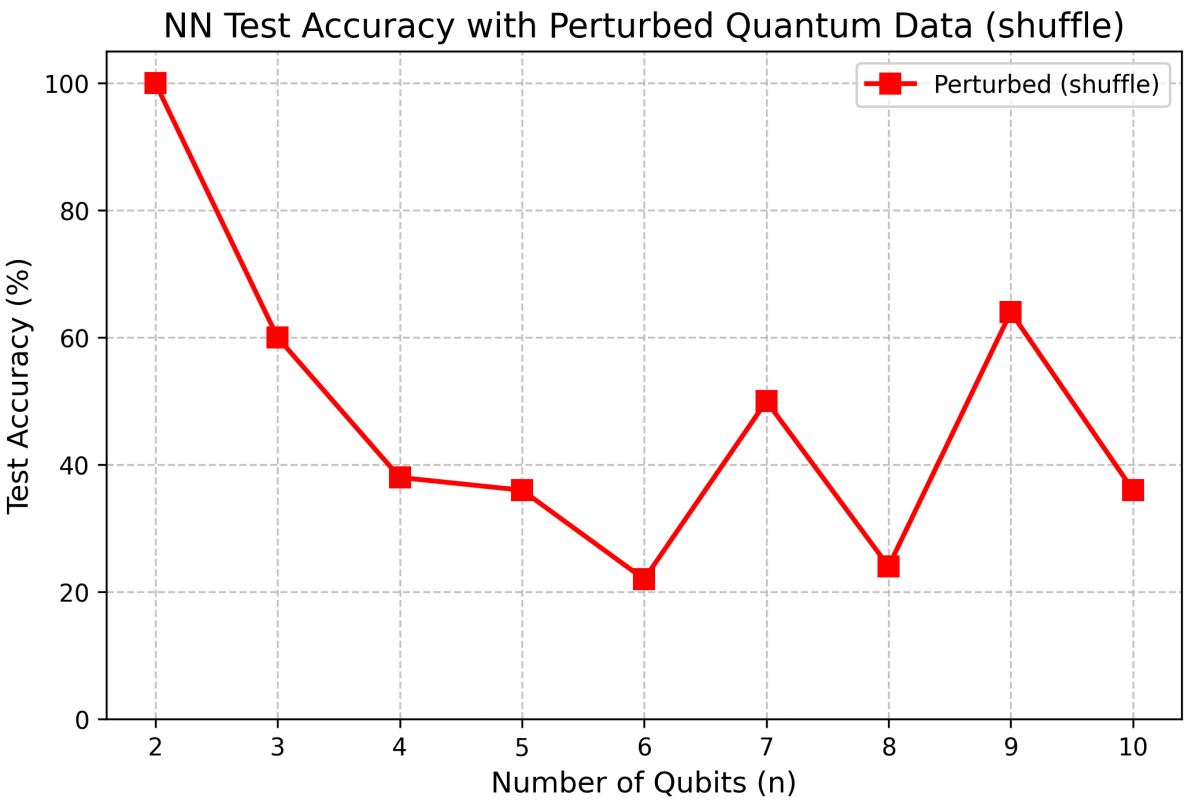
1. exponential size of nn
2. cost exponential resource to calculate r
3. size of counts, a and N do not match
4. the structure of nn is too simple

NN result



Ablation

to test if the nn actual use the quantum measurement result to calculate the result, instead of cheating: directly calculate r. We perturbate the measurement results. If the nn cheats, the results should not depend on the measurements.


NN Test Accuracy with Perturbed Quantum Data (shuffle)
Perturbed (shuffle)
Test Accuracy (%)
Number of Qubits (n)
0
20
40
60
80
100
2
3
4
5
6
7
8
9
10